Chapter 7E: Solid Oncology Overview

Cancer Screening, Staging, Treatment Modalities, Oncologic Emergencies, Treatment Toxicities, Survivorship, and Integrated Cases

This Medvale chapter is an original synthesis designed for internal medicine learners and future graph-RAG extraction. The uploaded hematology/oncology source provides coverage signals for oncology foundations, epidemiology, screening, staging, surgery, radiation, systemic therapy, breast cancer, plasma cell disorders, lymphomas, and leukemia introductions. This section expands the solid oncology and supportive-care framework with current guideline cross-checking.

Why this matters clinically

Cancer care appears in internal medicine in three ways: preventive care through vaccination, risk reduction, and screening; diagnosis and staging when a patient presents with a mass, abnormal imaging, cytopenias, hypercalcemia, thrombosis, or organ obstruction; and urgent management of complications from tumor burden or cancer therapy. The internist may not choose every chemotherapy regimen, but must recognize neutropenic sepsis, spinal cord compression, tumor lysis syndrome, hypercalcemia of malignancy, immune checkpoint inhibitor colitis, and paraneoplastic clues that reshape the differential.

This chapter treats oncology as a clinical reasoning framework rather than a catalog of every tumor. Individual cancers differ by organ and biology, but the same questions recur: Is the disease localized or metastatic? Is the intent curative or palliative? What emergency could kill the patient today? What toxicity is treatment-related? What should be screened, biopsied, staged, treated, monitored, or discussed as a goal of care?

Core illness scripts

Presentation pattern	Cancer-linked interpretation	Immediate danger to exclude
Weight loss, night sweats, anorexia, fatigue	Systemic inflammation, high tumor burden, lymphoma, metastatic tumor, occult infection	Sepsis, adrenal insufficiency, tuberculosis, uncontrolled diabetes, severe anemia
New focal bone pain, neurologic deficit, urinary retention	Bone metastasis or epidural spinal cord compression until proven otherwise	Spinal cord compression - emergency MRI and steroids when suspected
Cancer patient with fever, mucositis, ANC <500	Febrile neutropenia, catheter infection, pneumonia, enterocolitis, UTI	Septic shock; antibiotics should not wait for complete workup
Constipation, polyuria, confusion, shortened QT, Ca often >12	Hypercalcemia of malignancy from PTHrP, osteolysis, calcitriol excess	Arrhythmia, dehydration, AKI, coma
Bulky mediastinal mass with facial swelling, venous distention	Superior vena cava obstruction from lung cancer, lymphoma, thrombosis	Airway compromise, cerebral edema, hemodynamic compromise
After therapy: hyperK, hyperphos, high uric acid, low calcium	Tumor lysis syndrome in proliferative or bulky chemosensitive tumors	AKI, arrhythmia, seizure, sudden death
Diarrhea, hepatitis, pneumonitis, endocrinopathy after immunotherapy	Immune-related adverse event from checkpoint inhibitor	Colitis perforation, respiratory failure, adrenal crisis, myocarditis

Cancer biology and vocabulary for internists

A malignant tumor becomes clinically important because it can invade, metastasize, evade immune detection, sustain growth signaling, recruit blood supply, resist apoptosis, and damage host organs. Pathology vocabulary predicts management. Carcinoma arises from epithelial tissue; sarcoma from mesenchymal tissue; leukemia and lymphoma from hematolymphoid tissue; melanoma from melanocytes. Dysplasia means disordered growth without invasion, carcinoma in situ means malignant epithelial cells confined above the basement membrane, and invasive carcinoma means penetration into stroma with access to lymphatics, blood vessels, nerves, and tissue planes.

The practical distinction is not simply benign versus malignant. It is localized versus regionally advanced versus metastatic, and curable versus controllable versus primarily symptom-directed. A small colon cancer may be cured surgically. A bulky lymphoma may be curable even when disseminated. A pancreatic cancer may be unresectable because of vascular encasement even without distant metastasis. A prostate cancer may be biologically indolent despite malignancy. Graph-RAG nodes should therefore connect tumor type, stage, grade, biomarkers, symptoms, complications, and treatment intent.

Staging, grading, biomarkers, and treatment intent

TNM is the shared staging language for most solid tumors. T describes the primary tumor size or local invasion, N describes regional lymph node involvement, and M describes distant metastasis. Stage grouping converts TNM into stage I-IV categories, but the meaning differs by tumor. Grade describes microscopic aggressiveness and differentiation. Biomarkers refine prognosis and therapy selection: ER/PR/HER2 in breast cancer; EGFR, ALK, ROS1, BRAF, MET, RET, NTRK, KRAS, and PD-L1 in lung cancer; MSI/MMR status in colorectal cancer; PSA and Gleason Grade Group in prostate cancer; AFP in hepatocellular carcinoma; CA-125 in ovarian cancer; and BRAF in melanoma.

Term	What it asks	Clinical consequence
Stage	Where is the cancer anatomically?	Guides local vs systemic therapy and prognosis.
Grade	How abnormal or aggressive do cells look?	High grade often grows faster and needs more aggressive therapy.
Biomarker	What receptor or mutation drives the tumor?	Selects targeted therapy, immunotherapy, endocrine therapy, or resistance prediction.
Performance status	How functional is the patient?	Determines whether treatment is safe; often more predictive than age.
Treatment intent	Curative, disease-control, symptom-control, or end-of-life?	Clarifies toxicity tradeoffs and goals-of-care timing.

Cancer prevention and screening framework

Primary prevention reduces cancer incidence: avoid tobacco, reduce alcohol excess, vaccinate against HPV and hepatitis B, treat hepatitis C when present, reduce ultraviolet exposure, address obesity, and minimize occupational carcinogens. Secondary prevention detects premalignant or early malignant disease: colonoscopy removes adenomas, Pap/HPV testing detects cervical precancer, and mammography or low-dose CT can find earlier-stage breast or lung cancer. Tertiary prevention reduces recurrence and late toxicity through surveillance, lifestyle counseling, and survivorship care.

Cancer	Average-risk screening anchor	High-yield caveats
Breast	Biennial mammography age 40-74.	High genetic or familial risk may need earlier MRI-based strategy. Abnormal imaging requires diagnostic evaluation and biopsy logic.
Cervical	Age 21-29 cytology every 3 years; age 30-65 cytology every 3 years, hrHPV every 5 years, or cotesting every 5 years.	HPV vaccination prevents many cancers but does not eliminate screening need in current guidance.
Colorectal	Start at age 45 for average risk; continue through 75; individualized age 76-85.	Positive stool-based test requires colonoscopy. Iron deficiency anemia in men/postmenopausal women is GI blood loss until proven otherwise.
Lung	Annual low-dose CT age 50-80 with >=20 pack-years, current smoking or quit within 15 years.	Do not screen if unable or unwilling to undergo curative treatment. Smoking cessation gives the largest benefit.
Prostate	Shared decision-making, often age 55-69 for average-risk patients.	PSA can reduce metastatic disease but causes overdiagnosis and treatment harms.
Skin	No universal average-risk screening recommendation.	Counsel high-risk patients on UV protection and evaluate changing lesions using ABCDE/ugly-duckling signs.

Diagnostic approach to a suspected solid tumor

Approach to a mass, suspicious imaging finding, or cancer syndrome

1. Clarify tempo and danger: airway, neurologic deficit, sepsis, bleeding, obstruction, hypercalcemia, tumor lysis risk, severe pain, or organ failure require immediate stabilization.
2. Localize the problem anatomically with history, exam, labs, and imaging. Ask whether symptoms come from the primary tumor, metastasis, paraneoplastic physiology, or treatment toxicity.
3. Obtain tissue unless the diagnosis is already established and management can safely proceed. Choose the biopsy site that gives diagnosis and staging information with the least risk.
4. Stage before definitive therapy when staging changes management. CT, MRI, PET/CT, bone scan, brain MRI, or ultrasound are selected by tumor type.

5. Order tumor markers and biomarkers as decision tools, not as nonspecific screening tests. A tumor marker without context creates false positives and anxiety.
6. Determine treatment intent and patient fitness. A frail patient with metastatic disease needs symptom priorities and goals-of-care discussion as much as histology.

> Common diagnostic traps: empiric steroids before lymphoma biopsy, assuming a smoker with recurrent pneumonia only has infection when obstruction is present, forgetting pregnancy testing before teratogenic therapy, and ordering broad tumor-marker panels as screening tests.

Treatment modalities: what the internist must know

Cancer therapy is increasingly multimodal. Surgery removes localized disease and provides tissue. Radiation controls local/regional disease and can palliate pain, bleeding, brain metastases, SVC syndrome, or spinal cord compression. Cytotoxic chemotherapy attacks dividing cells but injures marrow, mucosa, hair follicles, gonads, and other proliferative tissues. Endocrine therapy deprives hormone-driven tumors of growth signals. Targeted therapy blocks driver pathways but creates pathway-specific toxicities. Immunotherapy releases antitumor immune responses but can inflame nearly any organ.

Modality	Typical role	Internist toxicity anchors
Surgery	Curative local control, staging, debulking, palliation of obstruction or bleeding.	Post-op VTE, infection, ileus, pain control, nutrition, functional decline, steroid management.
Radiation	Adjuvant, definitive local therapy, palliation of bone pain, brain metastasis, SVC, cord compression.	Fatigue, dermatitis, mucositis, pneumonitis, enteritis, cystitis, marrow suppression, late fibrosis.
Cytotoxic chemo	Systemic treatment for micrometastatic/metastatic disease; curative for some tumors.	Neutropenia, anemia, thrombocytopenia, nausea, mucositis, diarrhea, alopecia, neuropathy, infertility, organ toxicity.
Endocrine therapy	Breast/prostate and other hormone-sensitive cancers.	Hot flashes, VTE/endometrial risk with tamoxifen, osteoporosis/arthralgia with aromatase inhibitors, metabolic effects with androgen deprivation.
Targeted therapy	Driver mutation or pathway-selected therapy.	EGFR rash/diarrhea, VEGF hypertension/proteinuria, HER2 cardiomyopathy, QT/liver abnormalities.
Immunotherapy	Checkpoint inhibitors and cellular therapy.	Colitis, hepatitis, pneumonitis, endocrinopathies, nephritis, myocarditis; CAR-T can cause CRS/ICANS.

Chemotherapy and targeted therapy toxicity map

Drug/class clue	Classic toxicity pattern	Clinical association to remember
Anthracyclines	Dose-related cardiomyopathy, myelosuppression, vesicant injury.	Baseline cardiac risk matters; consider cumulative dose.
Bleomycin	Pneumonitis/fibrosis, skin changes.	New dyspnea needs evaluation; avoid unnecessary high oxygen exposure.
Cisplatin	Nephrotoxicity, ototoxicity, severe nausea, neuropathy, Mg/K wasting.	Hydration and electrolyte monitoring are high-yield.
Cyclophosphamide/ifosfamide	Hemorrhagic cystitis, myelosuppression, infertility, SIADH in some settings.	Mesna protects bladder from acrolein.
Taxanes	Peripheral neuropathy, hypersensitivity, myelosuppression, fluid retention.	Stocking-glove neuropathy may impair gait and falls.
Vincristine	Neuropathy, ileus/constipation, SIADH; little myelosuppression.	Never intrathecal; autonomic neuropathy causes severe constipation.
Antimetabolites	Mucositis, myelosuppression, organ-specific toxicities.	Methotrexate uses leucovorin rescue; 5-FU/capecitabine can cause hand-foot and cardiotoxicity.
Checkpoint inhibitors	Immune colitis, hepatitis, pneumonitis, endocrinopathy, myocarditis.	Toxicity can occur after treatment stops; steroids treat many moderate-severe events after infection is excluded.

Oncologic emergencies: recognition-first algorithms

Oncologic emergencies are defined by time-sensitive organ threat rather than cancer type. Recognize the syndrome, stabilize first, notify oncology early, and avoid waiting for perfect staging before treating life-threatening physiology.

Emergency	Typical clues	Initial internal medicine actions
Febrile neutropenia	Fever with ANC <500/uL or expected nadir; mucositis, catheter, pneumonia, abdominal pain, hypotension.	Cultures but do not delay antibiotics; antipseudomonal beta-lactam within 1 hour; risk stratify inpatient vs outpatient.
Hypercalcemia	Confusion, constipation, polyuria, dehydration, AKI, QT shortening; Ca often >12 and severe >14.	IV isotonic fluids if not overloaded, calcitonin bridge, IV bisphosphonate or denosumab, treat tumor.
Spinal cord compression	Back pain worse lying down, focal weakness/sensory level, bowel/bladder dysfunction.	Immediate MRI, dexamethasone if suspected, neurosurgery/radiation oncology; protect ambulation.
SVC syndrome	Face/neck/arm swelling, venous distention, cough, dyspnea, worse bending forward.	Assess airway/cerebral edema; CT chest if stable; tissue diagnosis if unknown; stent/radiation/chemo based on severity.
Tumor lysis syndrome	High-risk tumor or recent therapy; hyperK, hyperphos, hypocalcemia, hyperuricemia, AKI.	IV fluids if safe, telemetry, frequent labs, allopurinol prophylaxis or rasburicase treatment, dialysis if refractory.
Hyperviscosity	Headache, visual changes, mucosal bleeding, neurologic symptoms; IgM or myeloma.	Urgent plasma exchange for symptoms; avoid RBC transfusion before viscosity control unless essential.
Malignant tamponade	Dyspnea, tachycardia, hypotension, JVD, pulsus paradoxus.	Bedside echo, urgent drainage if tamponade, cytology, recurrence planning.
Checkpoint toxicity	Dyspnea, diarrhea, transaminitis, endocrinopathy after immunotherapy.	Hold immunotherapy, grade severity, exclude infection/ACS/PE, early steroids for moderate-severe toxicity.

Cancer-associated thrombosis and bleeding

Cancer produces both thrombosis and bleeding. Thrombosis is driven by tumor procoagulants, inflammation, immobility, surgery, central venous catheters, chemotherapy, antiangiogenic therapy, and direct vessel compression. Mucinous adenocarcinomas, pancreatic cancer, gastric cancer, lung cancer, ovarian cancer, brain tumors, and advanced metastatic disease are especially thrombogenic. Bleeding occurs from tumor erosion, thrombocytopenia, DIC, liver dysfunction, anticoagulants, renal failure, and mucosal injury.

The practical question is whether the anticoagulant plan is safe given platelet count, active bleeding, brain metastases, renal function, drug interactions, procedures, GI/GU mucosal tumors, and goals of care. Low-molecular-weight heparin and direct oral anticoagulants are commonly used, but GI/GU bleeding risk and drug interactions may favor one approach over another. IVC filters are reserved for selected acute VTE patients with a temporary contraindication to anticoagulation, not routine add-ons.

High-yield solid tumor illness scripts

Tumor	Presentation clues	Internal medicine links
Lung cancer	Cough, hemoptysis, weight loss, recurrent pneumonia, smoking history, lung nodule; SVC or brain/bone/adrenal metastases.	Screen selected high-risk adults with LDCT. Small-cell links to SIADH, ectopic ACTH, Lambert-Eaton; squamous links to PTHrP hypercalcemia.
Breast cancer	Screen-detected or palpable hard irregular mass, nipple retraction/discharge, axillary nodes, bone pain.	ER/PR/HER2 guide therapy. Watch bone metastases, lymphedema, anthracycline/HER2 cardiotoxicity.
Colorectal cancer	Iron deficiency anemia, occult blood, change in stool caliber, obstruction, weight loss.	Screening prevents cancer by polyp removal. Right-sided lesions cause anemia; left-sided lesions cause obstruction/change in stool.
Prostate cancer	PSA-detected; urinary symptoms often BPH; bone pain from osteoblastic metastases.	Shared decision-making for PSA. Androgen deprivation causes hot flashes, metabolic syndrome, osteoporosis, sarcopenia.
Pancreatic cancer	Painless jaundice, weight loss, epigastric/back pain, new diabetes, migratory thrombophlebitis.	Often late. Manage biliary obstruction/cholangitis, cachexia, VTE, pain, nutrition.
Hepatocellular carcinoma	Cirrhosis, HBV/HCV, MASLD/MASH, alcohol; RUQ pain, weight loss, decompensation.	Surveillance in high-risk cirrhosis uses ultrasound +/- AFP. Liver reserve drives therapy feasibility.
Ovarian cancer	Bloating, early satiety, pelvic/abdominal pain, ascites, adnexal mass.	Often late-stage; CA-125 is not a general screening test; BRCA/Lynch affect family risk.
Melanoma	Changing pigmented lesion, ABCDE features, nodal disease; brain/lung/liver metastases.	BRAF and immunotherapy relevant; immune-related adverse events common with checkpoint inhibitors.

Paraneoplastic syndromes and tumor clues

Syndrome	Classic tumor association	Mechanistic/clinical clue
SIADH	Small-cell lung cancer, CNS tumors, pulmonary disease	Euvolemic hypotonic hyponatremia with concentrated urine.
PTHrP hypercalcemia	Squamous lung/head-neck, renal, bladder, ovarian	High calcium, low PTH, often normal/low vitamin D.
Ectopic ACTH	Small-cell lung cancer, bronchial carcinoid	Cushing physiology, hypokalemic alkalosis, hyperglycemia, infections.
Lambert-Eaton	Small-cell lung cancer	Proximal weakness improves with repeated use; autonomic symptoms.
Trousseau syndrome	Pancreatic, gastric, lung, mucinous adenocarcinoma	Recurrent migratory thrombosis; consider occult malignancy.
Dermatomyositis	Ovarian, lung, pancreatic, stomach, colorectal, others	Proximal weakness with heliotrope rash/Gottron papules.
Acanthosis nigricans / Leser-Trelat	Gastric and other adenocarcinomas	Abrupt severe skin findings in older adult can be paraneoplastic.

Supportive care, palliative care, and survivorship

Supportive care is concurrent care that preserves treatment options and prevents avoidable admissions. Nausea, constipation, diarrhea, mucositis, anorexia/cachexia, pain, depression, insomnia, neuropathy, cytopenias, sexual dysfunction, fertility concerns, financial toxicity, and caregiver strain should be actively screened. Palliative care can occur alongside disease-directed therapy when symptoms are high, prognosis is limited, decision-making is complex, or care goals are changing.

Survivorship requires a treatment summary and risk-based plan. Patients may need surveillance for recurrence, management of cardiomyopathy after anthracyclines or HER2 therapy, thyroid disease after neck radiation, pulmonary fibrosis after bleomycin or thoracic radiation, neuropathy after platinum/taxane/vinca therapy, premature menopause or infertility, osteoporosis after endocrine therapy, second malignancy risk, cognitive symptoms, return-to-work support, vaccination review, and cardiovascular risk reduction.

Original diagnostic diagrams

Diagram 1. First-pass sorting of a cancer-related presentation

Patient clue	Sort first by threat	Then link to likely mechanism
Fever after chemotherapy	Sepsis risk	Febrile neutropenia -> cultures, antibiotics within 1 hour, risk stratification.
Back pain + weakness/bladder symptoms	Neurologic preservation	Epidural spinal cord compression -> MRI, steroids, decompression/radiation.
Confusion + constipation + AKI	Metabolic emergency	Hypercalcemia -> fluids, calcitonin, antiresorptive therapy, ECG/renal monitoring.
Facial swelling + dyspnea	Airway/venous return	SVC syndrome -> airway assessment, CT, stent/radiation/chemo.
New diarrhea on checkpoint inhibitor	Inflammatory toxicity	Immune colitis vs infection -> grade, stool studies, steroids if moderate-severe.
Anemia + rouleaux + renal injury	Plasma cell disorder	Myeloma physiology -> SPEP/UPEP/free light chains, calcium, renal support.

Diagram 2. Treatment intent ladder

Intent	Typical examples	What the internist should ask
Prevention	HPV/HBV vaccination, tobacco cessation, colon polyp removal.	Is the patient eligible and up to date? What risk factor can be changed today?
Curative local therapy	Early breast, colon, prostate, lung, melanoma surgery/radiation.	Are staging and pre-op medical optimization complete?
Curative systemic therapy	Testicular cancer, many lymphomas/leukemias, selected adjuvant regimens.	Can the patient tolerate curative toxicity? How will neutropenia be handled?
Disease control	Metastatic hormone-sensitive or targetable cancers.	What is the biomarker? What toxicity should be monitored?
Symptom-directed palliation	Painful bone metastasis, obstruction, bleeding, dyspnea.	What symptom matters most and what intervention works fastest?
End-of-life care	Progressive disease despite options or unacceptable toxicity.	Are code status, hospice, caregiver support, and medication simplification addressed?

Integrated cases

Case 1 - Fever after chemotherapy: A 62-year-old with breast cancer presents 8 days after chemotherapy with T 38.6 C, ANC 300/uL, mucositis, and BP 96/58. The key diagnosis is febrile neutropenia with possible sepsis. Draw cultures and evaluate likely sources, but administer empiric antipseudomonal antibiotics immediately. Inpatient management is favored by hypotension and mucositis.

Case 2 - Back pain with urinary retention: A man with metastatic prostate cancer has new thoracic back pain, leg weakness, and urinary retention. Treat as spinal cord compression. Give corticosteroids if no contraindication, obtain urgent MRI, and consult neurosurgery/radiation oncology. Preserving ambulation is time-sensitive.

Case 3 - Hypercalcemia with renal injury: A patient with squamous cell lung cancer has calcium 14.2 mg/dL, creatinine 2.1, constipation, and confusion. Initial therapy is isotonic fluid if tolerated, calcitonin for rapid short-term lowering, and IV bisphosphonate or denosumab for sustained control.

Case 4 - Screening versus diagnostic testing: A 52-year-old average-risk adult asks about colon cancer screening. This is screening, so options and intervals can be discussed. A 52-year-old with iron deficiency anemia and positive FIT needs diagnostic colonoscopy; repeating stool testing is inappropriate.

Case 5 - Immunotherapy diarrhea: A melanoma patient on pembrolizumab develops six watery stools per day and abdominal pain. Consider immune-mediated colitis, but exclude infection including C. difficile when appropriate. Hold immunotherapy and treat moderate-severe immune toxicity with systemic corticosteroids after grading.

Case 6 - New jaundice and weight loss: Painless jaundice, pale stools, dark urine, weight loss, and a distended gallbladder suggest malignant biliary obstruction. Priorities are cholangitis assessment, bilirubin/coagulation/nutrition status, decompression if indicated, and tissue/staging coordination.

Common pitfalls

- Treating all weight loss as cancer without checking mimics such as depression, hyperthyroidism, diabetes, chronic infection, medication effects, and substance use.
- Ordering broad tumor marker panels as screening tests; tumor markers usually need a specific clinical question.
- Delaying antibiotics in febrile neutropenia while waiting for imaging or urine results.
- Missing spinal cord compression because the first complaint is pain rather than paralysis.
- Giving high-volume IV fluids for hypercalcemia or TLS without reassessing heart failure, kidney function, urine output, and oxygenation.
- Assuming immune checkpoint toxicity cannot occur because the last infusion was weeks earlier.
- Forgetting that cancer patients can have ordinary diseases: ACS, PE, pneumonia, gallstones, medication adverse effects, and endocrine disease remain common.
- Equating palliative care with giving up; palliative care can improve symptom control while disease-directed therapy continues.

Practice questions

1. A patient on chemotherapy has T 38.4 C and ANC 200/uL. Most important first principle? Answer: obtain cultures and give empiric antibacterial therapy promptly, generally within 1 hour; risk stratification determines inpatient versus carefully selected outpatient management.

2. A patient with squamous lung cancer has calcium 14.8 mg/dL, confusion, constipation, polyuria, and AKI. Treatment sequence? Answer: IV isotonic fluids if tolerated, calcitonin for rapid temporary lowering, antiresorptive therapy such as IV bisphosphonate or denosumab, and treatment of the malignancy.

3. What symptom combination should trigger concern for malignant spinal cord compression? Answer: new severe back pain in a cancer patient, especially with weakness, sensory level, gait impairment, bowel/bladder dysfunction, or pain worse when lying down.

4. Why is a positive FIT/stool DNA test not repeated before colonoscopy? Answer: a positive screening test becomes a diagnostic problem; colonoscopy is required to identify or remove neoplasia.

5. A patient on checkpoint inhibitor therapy develops dyspnea and bilateral ground-glass opacities. What must be considered? Answer: immune-mediated pneumonitis, while evaluating infection, PE, edema, and progression; moderate-severe cases require holding therapy and corticosteroid-based management.

6. A smoker has a central lung mass, hyponatremia with concentrated urine, and low serum osmolality. Likely syndrome? Answer: SIADH, classically associated with small-cell lung cancer.

7. Which cancers have high-yield population screening programs with major U.S. recommendations? Answer: breast, cervical, colorectal, and lung cancer in defined groups; prostate screening is individualized/shared decision-making.

8. A bulky lymphoma patient starts chemotherapy and develops hyperK, hyperphos, hypocalcemia, high uric acid, and rising creatinine. Diagnosis and management? Answer: tumor lysis syndrome; manage with IV fluids if safe, telemetry, electrolyte treatment, urate-lowering therapy such as rasburicase when indicated, and dialysis for refractory abnormalities.

Graph-RAG extraction map

The extraction map below is formatted so a future Medvale graph-RAG pipeline can create nodes, edges, and retrieval hooks. Include syndromes, labs, treatments, toxicities, screening tests, and emergency actions as nodes.

Node	Class / subclass	High-value edges
Solid oncology overview	Discipline framework -> oncology	connects_to: cancer screening, TNM staging, treatment intent, oncologic emergencies, survivorship
TNM staging	Diagnostic framework	determines: surgery/radiation/systemic therapy; interacts_with: biomarkers, grade, performance status
Cancer screening	Preventive medicine	includes: breast, cervical, colorectal, lung; modified_by: age, risk, prior results, life expectancy
Febrile neutropenia	Oncologic emergency / infectious disease	caused_by: chemotherapy marrow suppression; treated_by: rapid antipseudomonal antibiotics; risk_stratified_by: ANC, stability, logistics
Hypercalcemia of malignancy	Oncologic emergency / electrolyte	caused_by: PTHrP, bone metastases, calcitriol; treated_by: fluids, calcitonin, bisphosphonate/denosumab
Spinal cord compression	Oncologic emergency / neurologic	presents_with: back pain, weakness, bladder dysfunction; diagnosed_by: MRI; treated_by: steroids, surgery/radiation
Tumor lysis syndrome	Oncologic emergency / renal-electrolyte	causes: hyperK, hyperphos, hypocalcemia, hyperuricemia, AKI; prevented_by: hydration, allopurinol/rasburicase
Immune-related adverse events	Treatment toxicity / immunology	caused_by: checkpoint inhibitors; affects: colon, lung, liver, endocrine glands, kidney, heart; treated_by: hold drug, steroids by grade
Cancer-associated thrombosis	Hemostasis / oncology	caused_by: tumor inflammation, catheters, chemotherapy; treated_by: anticoagulation selected by bleeding risk
Palliative care	Care model	improves: symptoms, decision quality; coexists_with: disease-directed treatment when needs are high

Reference anchors used for clinical cross-checking

- Uploaded source coverage: heme3.pdf oncology foundations, epidemiology, cancer screening, staging/therapy, breast cancer, plasma cell disorders, lymphomas, and leukemia introduction.
- USPSTF A and B recommendations: breast cancer screening age 40-74 biennial; cervical screening age 21-65 strategies; colorectal screening age 45-75; lung cancer annual LDCT age 50-80 with >=20 pack-year history and current smoking or quit within 15 years.
- CDC cancer screening overview: routine screening is emphasized for breast, cervical, colorectal, and lung cancers in recommended populations.

- NCI treatment adverse-effect resources: anemia, thrombocytopenia, infection/neutropenia, diarrhea, neuropathy, organ-related inflammation and immunotherapy, and other treatment toxicities.
- ASCO/IDSA febrile neutropenia outpatient management guideline: empiric antibacterial therapy within 1 hour of triage and monitoring/risk stratification for selected outpatient care.
- Alberta Health Services 2025 oncologic emergencies guideline resource: spinal cord compression, febrile neutropenia, hypercalcemia, hyperviscosity, checkpoint inhibitor toxicities, severe SIADH, and tumor lysis syndrome as emergency categories.